Enzyme Classification and Nomenclature

Chapter · April 2001

DOI: 10.1038/npg.els.0000710

Sinéad Boyce
Trinity College Dublin

Enzyme Classification and Nomenclature

Introductory article

Article Contents

Sinead Boyce, *Trinity College, Dublin, Ireland*

Keith F Tipton, *Trinity College, Dublin, Ireland*

Enzyme classification and nomenclature is a system that allows the unambiguous identification of enzymes in terms of the reactions they catalyse. This relies on a numerical system to class enzymes in groups according to the types of reaction catalysed and systematic naming that describes the chemical reaction involved.

Introduction

The need for a rational nomenclature for enzymes can be seen from the plethora of unhelpful names for enzymes in the earlier literature. Only those who were directly involved might have known the difference between the old yellow enzyme and the new yellow enzyme and what diaphorase, or for that matter DT-diaphorase, catalysed (try EC 1.6.99.1 and EC 1.8.1.4). Similarly, the reaction catalysed by rhodenese (thiosulfate sulfurtransferase: EC 2.8.1.1) was not apparent from its name and use of the name urokinase to describe a peptidase (EC 3.4.21.73) is confusing since the term kinase is usually used for enzymes that transfer phosphate from ATP to another substrate.

In trying to bring some order to the chaotic situation of enzyme nomenclature, Malcolm Dixon and Edwin Webb, in 1958, took a step that was radically different from that used in other branches of nomenclature by classifying enzymes in terms of the reactions they catalysed, rather than by their structures. This system has been adopted and developed by the International Union of Biochemistry and Molecular Biology (IUBMB), through its Joint Nomenclature Committee with the International Union of Pure and Applied Chemistry (IUPAC) into the *Enzyme Nomenclature* list of enzymes. This has been through several editions, the most recent being published in 1992. This material is also made available in a modified form through the SWISSPROT ENZYME on-line database. More recent additions and modifications to the list have been published as supplements in the *European Journal of Biochemistry* and the most recent additions are available on-line (see Enzyme Supplement, 1999 in Further Reading section).

Detailed rules for naming and classifying enzymes have been formulated and it is a relatively easy matter to assign an enzyme to an overall class and to give it a name that describes what it does. For example, if it oxidizes something by reducing NAD(P), it is a dehydrogenase classified as EC 1. x. 1. – , where the number x refers to the group oxidized: 1 for –CHOH–, 2 for aldehyde or ketone, etc.; however, if it transfers a phosphate, diphosphate or another phosphate-containing group through its phosphate to another substrate, it is a phosphotransferase classified as EC 2.7. z.–, where z refers to the nature of the acceptor group, and so on. The detailed descriptions of the procedures for assigning enzymes to specific classes and subclasses and the rules for systematic enzyme names that have been approved by the IUBMB Nomenclature Committee have been published in *Enzyme Nomenclature* (1992). The account below has been adapted from the fuller material in that source, which should be consulted if further detail is required.

General Classification Structure

The basic layout of the classification for each enzyme is described below with some indication of the guidelines followed. Further details of the principles governing the nomenclature of individual enzyme classes are given in the following sections.

EC number: The classification number, which is made up of four digits, identifies the enzyme by the reaction catalysed. This is also valuable for relating the information to other databases.

Recommended name: The most commonly used name for the enzyme is usually used, provided that it is unambiguous. A number of generic words indicating reaction types may be used in recommended names, but not in the systematic names, e.g. *dehydrogenase*, *reductase*, *oxidase*, *peroxidase*, *kinase*, *tautomerase*, *deaminase*, *dehydratase*, etc. Where additional information is needed to make the reaction clear, a phrase indicating the reaction or a product may be added in parentheses after the second part of the name, e.g. (*ADP-forming*), (*dimerizing*), (*CoA-acylating*).

Reaction: The actual reaction catalysed, written, where possible, in the form of a 'biochemical' equation [I]:

$$A + B = P + Q \qquad [I]$$

This formulation gives no indication of the preferred equilibrium of the reaction or, indeed, whether it is readily reversible. In the case of reversible reactions, the direction chosen for the reaction and systematic name is the same for all the enzymes in a given class, even if this direction has not been demonstrated for all. Thus, systematic names are based on this written reaction, even though only the reverse of this may actually have been demonstrated experimentally. Frequently, such biochemical equations are neither charge- nor mass-balanced.

Other name(s): Any other names that have been used for the enzyme. This is to be as comprehensive a list as possible to aid searching for any specific enzyme. The inclusion of a name in this list does not mean that its use is encouraged. In some cases where the same name has been given to more than one enzyme, this ambiguity will be indicated.

Systematic name: This attempts to describe in unambiguous terms what the enzyme actually catalyses. Systematic names consist of two parts. The first contains the name of the substrate or, in the case of a bimolecular reaction, of the two substrates separated by a colon. The second part, ending in *-ase*, indicates the nature of the reaction. A number of generic words indicating a type of reaction may be used in either recommended or systematic names: *oxidoreductase*, *oxygenase*, *transferase* (with a prefix indicating the nature of the group transferred), *hydrolase*, *lyase*, *racemase*, *epimerase*, *isomerase*, *mutase*, *ligase*. Where additional information is needed to make the reaction clear, a phrase indicating the reaction or a product should be added in parentheses after the second part of the name, e.g. (*ADP-forming*), (*dimerizing*), (*CoA-acylating*).

Comments: Brief comments on the nature of the reaction catalysed, possible relationships to other enzymes, species differences, metal-ion requirement, etc.

References: Key references on the identification, nature, properties and function of the enzyme. These have been omitted from the examples below to save space.

Notes on Chemical Nomenclature

Although a detailed description of chemical nomenclature is beyond the scope of this article, some comments are necessary because the fearsome names used are often difficult for a biochemist to understand. The aim of the chemist is to be able to name a compound in such a way that anyone who knows the rules of chemical nomenclature can write down its chemical structure and formula from it. Therefore, it must be unambiguous both in terms of all the chemical groups that make up the compound, how and where they are linked together and the compound's stereochemistry. This does lead to names that are not much help for general use, for example, the neurotransmitter norepinephrine (noradrenaline) has a systematic name of (*R*)-4-(2-amino-1-hydroxyethyl)-1,2-benzenediol, the antibiotic benzylpenicillin is [2*S*-(2α,5α, 6β)]-3,3-dimethyl-7-oxo-6-[(phenylacetyl)amino]-4-thia-1-azabicyclo[3.2.0]heptane-2-carboxylic acid and aspirin is 2-(acetyloxy)benzoic acid.

The basic rules for writing down the systematic name of a compound are to take a basic, or root, structure or its derivative, for example benzoic acid is the derivative of the root benzene in the case of aspirin. The substituents are then written before it with the position of each substituent and any stereochemistry being identified. There are several possible modifications of this procedure and it is possible to write more than one systematic name that is more-or-less unambiguous (see for example, the alternative names that have been used for norepinephrine in the Merck Index). Variations arise, for example, from the choice of root compound and the order in which the substituents are written. Where systematic names are used, the enzyme classification system uses the IUPAC system, which uses rather few root compounds and writes the substituents in alphabetical order (e.g. amino before hydroxy, before methyl etc.). Chemists use fewer root structures than biochemists. For example, biochemists all know the amino acid tryptophan and that it can be decarboxylated to tryptamine. Therefore, they have no trouble with naming the hormone melatonin *N*-acetyl-5-methoxytryptamine. However, if a chemist does not accept tryptamine as a root structure it would become *N*-[2-(5-methoxy-1*H*-indol-3-yl)ethyl]acetamide. Because the enzyme classification system is primarily designed for biochemists, the biochemical names are frequently used, where these are widely known. However, collaboration with IUPAC ensures that the systematic names can be readily found from these in their literature.

It should be noted that the systematic names of norepinephrine and melatonin are single 'words', which can contain lots of hyphens in them and generally systematic names are written as single 'words'. Among the few exceptions to this general rule are acids, including phosphates, as shown by the examples of penicillin and aspirin, where 'acid' is written as a separate word. Similarly creatine phosphate is written as two words, although it is possible to write the compound as a single word by rearranging the name to phosphocreatine. An example of such rearrangement is the name 6-phosphofructokinase for the enzyme (EC 2.7.1.11) that catalyses the reaction:

$$\text{fructose 6-phosphate} + \text{ATP} = \text{fructose 1,6-bisphosphate} + \text{ADP}$$

Note that the term bisphosphate is used here rather than diphosphate. In order to avoid confusion, diphosphate is used only for cases where the two phosphates are linked together (as in adenosine diphosphate; ADP) whereas bisphosphates have the two phosphates attached to separate groups in the molecule.

When a substrate name has two words there is a potential problem using them in enzyme names. Glucose-6-phosphate 1-dehydrogenase (EC 1.1.1.49) catalyses the reaction:

D-glucose 6-phosphate + NADP = D-glucono-1,5-lactone 6-phosphate + $NADPH_2$

but in order to indicate that the substrate oxidized is glucose 6-phosphate, not just phosphate, an extra hyphen is added to the enzyme name.

In denoting stereochemistry, the IUPAC rules prefer the *R*- and *S*- system and this is generally used for enzyme nomenclature. However, in the case of sugars and amino acids, the D- and L- designations are so well known that they are followed in the enzyme list. The use of italics in chemical names can at first seem rather odd, but the simplest way of thinking about it is to think how one would look up the name of a compound in an index, for example *N*-acetyl-5-methoxytryptamine would be found by searching through A for acetyl not N for *N*-acetyl. Clearly the same applies to *R*- and *S*-isomers. Therefore, the italic can be taken to mean 'do not bother to look under this letter in any index'. Having adopted this way of doing things, it is logical also to use italics for these when they occur in the middle of a name. The exceptions to this general rule are the D- and L- designations, which are not italicized, however, these are written, by convention, in a smaller size than normal.

The above summary glosses over many of the complexities of systematic chemical nomenclature and fuller details of the rules and their application can be found in *A Guide to IUPAC Nomenclature of Organic Compounds* (1994) .

Enzyme Classes and Definitions

Class 1. –.–.– Oxidoreductases

This class contains the enzymes catalysing oxidation reactions. Since the oxidation of one group must be accompanied by the reduction of another, they are grouped together as oxidoreductases. The systematic enzyme name is in the form *donor:acceptor oxidoreductase*. The substrate that is being oxidized is regarded as being the hydrogen donor. The recommended name is commonly *donor dehydrogenase*. Although the term *reductase* is sometimes used as an alternative, it is important to remember that the recommended name does not define the equilibrium position of the reaction or the net direction of flux through the enzyme *in vivo*. Indeed, in some cases, an enzyme within a metabolic pathway can proceed in a thermodynamically unfavoured direction because of the effective removal of one of the reactants. The term *donor oxidase* is used only when O_2 is the acceptor.

The second figure in the code number of the oxidoreductases denotes the type of group in the hydrogen-donor substrate that is oxidized or reduced. The third number denotes the hydrogen acceptor: 1 denotes NAD(P), 2 a cytochrome, 3 molecular oxygen, 4 a disulfide, 5 a quinone or similar compound, 6 a nitrogenous group, 7 an iron–sulfur protein and 8 a flavin. The number 99 is used for all other acceptors. This group contains a number of enzymes that have been shown to work with synthetic acceptors, such as 2,6-dichloroindophenol or phenazine methosulfate, but where the physiological acceptor is unknown. It is intended that they should be transferred to more descriptive sub-subclasses when the natural acceptor is identified.

For subclasses 1.13 and 1.14, a different classification scheme is used since these enzymes catalyse the incorporation of oxygen into the substrate. The recommended names are generally monooxygenase or dioxygenase, depending on whether one or two atoms of oxygen are incorporated into the substance oxidized. The sub-subclasses are numbered from 11 onwards.

Table 1 summarizes the structure of Class 1.

Examples

EC 1.1.1.14

Recommended name: L-iditol 2-dehydrogenase

Reaction: L-iditol + NAD = L-sorbose + $NADH_2$

Other name(s): polyol dehydrogenase; sorbitol dehydrogenase

Systematic name: L-iditol:NAD 2-oxidoreductase

Comments: Also acts on D-glucitol (giving D-fructose) and other closely related sugar alcohols.

EC 1.14.13.59

Recommended name: L-lysine 6-monooxygenase ($NADPH_2$)

Reaction: L-lysine + $NADPH_2$ + O_2 = N^6-hydroxy-L-lysine + NADP + H_2O

Other name(s): lysine N^6-hydroxylase

Systematic name: L-lysine, $NADPH_2$:oxygen oxidoreductase (6-hydroxylating)

Comments: A flavoprotein (FAD). The enzyme from strain EN 222 of *E. coli* is highly specific for L-lysine; L-ornithine and L-homolysine are, for example, not substrates. A lysine monooxygenase (EC 1.13.12.10) from this organism has been reported to catalyse the same hydroxylation without the involvement of NAD(P)H_2.

Class 2.–.–.– Transferases

These enzymes transfer a group from one substrate (the donor) to another (the acceptor) according to the general reaction [II]:

$$X–Y + Z = X + Y–Z \qquad [II]$$

Table 1 Class 1.–.–.– Oxidorectuses

1. 1.–.– Acting on the CH–OH group of donors	
	1. 1. 1.– With NAD or NADP as acceptor
	1. 1. 2.– With a cytochrome as acceptor
	1. 1. 3.– With oxygen as acceptor
	1. 1. 4.– With a disulfide as acceptor
	1. 1. 5.– With a quinone or similar compound as acceptor
	1. 1.99. – With other acceptors
1. 2. –.– Acting on the aldehyde or oxo group of donors	
	1. 2. 1.– With NAD or NADP as acceptor
	1. 2. 2.– With a cytochrome as acceptor
	1. 2. 3.– With oxygen as acceptor
	1. 2. 4.– With a disulfide as acceptor
	1. 2. 7.– With an iron–sulfur protein as acceptor
	1. 2.99.– With other acceptors
1. 3.–.– Acting on the CH–CH group of donors	
	1. 3. 1.– With NAD or NADP as acceptor
	1. 3. 2.– With a cytochrome as acceptor
	1. 3. 3.– With oxygen as acceptor
	1. 3. 5.– With a quinone or related compound as acceptor
	1. 3. 7.– With an iron–sulfur protein as acceptor
	1. 3.99.– With other acceptors
1. 4.–.– Acting on the CH–NH_2 group of donors	
	1. 4. 1.– With NAD or NADP as acceptor
	1. 4. 2.– With a cytochrome as acceptor
	1. 4. 3.– With oxygen as acceptor
	1. 4. 4.– With a disulfide as acceptor
	1. 4. 7.– With an iron–sulfur protein as acceptor
	1. 4.99.– With other acceptors
1. 5.–.– Acting on the CH–NH group of donors	
	1. 5. 1.– With NAD or NADP as acceptor
	1. 5. 3.– With oxygen as acceptor
	1. 5. 4.– With a disulfide as acceptor
	1. 5. 5.– With a quinone or similar compound as acceptor
	1. 5.99.– With other acceptors
1. 6.–.– Acting on $NADH_2$ or $NADPH_2$	
	1. 6. 1.– With NAD or NADP as acceptor
	1. 6. 2.– With a cytochrome as acceptor
	1. 6. 4.– With a disulfide as acceptor
	1. 6. 5.– With a quinone or similar compound as acceptor
	1. 6. 6.– With a nitrogenous group as acceptor
	1. 6. 8.– With a flavin as acceptor
	1. 6.99.– With other acceptors
1. 7.–.– Acting on other nitrogenous compounds as donors	
	1. 7. 2.– With a cytochrome as acceptor
	1. 7. 3.– With oxygen as acceptor
	1. 7. 7.– With an iron–sulfur protein as acceptor
	1. 7.99.– With other acceptors

continued

Table 1 – *continued*

1. 8.–.– Acting on a sulfur group of donors
    1. 8. 1.– With NAD or NADP as acceptor
    1. 8. 2.– With a cytochrome as acceptor
    1. 8. 3.– With oxygen as acceptor
    1. 8. 4.– With a disulfide as acceptor
    1. 8. 5.– With a quinone or similar compound as acceptor
    1. 8. 7.– With an iron–sulfur protein as acceptor
    1. 8.99.– With other acceptors
1. 9.–.– Acting on a haem group of donors
    1. 9. 3.– With oxygen as acceptor
    1. 9. 6.– With a nitrogenous group as acceptor
    1. 9.99.– With other acceptors
1.10.–.– Acting on diphenols and related substances as donors
    1.10. 1.– With NAD or NADP as acceptor
    1.10. 2.– With a cytochrome as acceptor
    1.10. 3.– With oxygen as acceptor
    1.10.99.– With other acceptors
1.11.–.– Acting on a peroxide as acceptor (peroxidases)
    1.11.1.– A single subclass containing the peroxidases
1.12.–.– Acting on hydrogen as donor
    1.12. 1.– With NAD or NADP as acceptor
    1.12. 2.– With a cytochrome as acceptor
    1.12.99.– With other acceptors
1.13.–.– Acting on single donors with incorporation of molecular oxygen
    1.13.11.– With incorporation of two atoms of oxygen
    1.13.12.– With incorporation of one atom of oxygen
    1.13.99.– Miscellaneous (requires further characterization)
1.14.–.– Acting on paired donors with incorporation of molecular oxygen
    1.14.11.– With 2-oxoglutarate as one donor, and incorporation of one atom each of oxygen into both donors
    1.14.12.– With $NADH_2$ or $NADPH_2$ as one donor, and incorporation of two atoms of oxygen into one donor
    1.14.13.– With $NADH_2$ or $NADPH_2$ as one donor, and incorporation of one atom of oxygen
    1.14.14.– With reduced flavin or flavoprotein as one donor, and incorporation of one atom of oxygen
    1.14.15.– With a reduced iron–sulfur protein as one donor, and incorporation of one atom of oxygen
    1.14.16.– With reduced pteridine as one donor, and incorporation of one atom of oxygen
    1.14.17.– With ascorbate as one donor, and incorporation of one atom of oxygen
    1.14.18.– With another compound as one donor, and incorporation of one atom of oxygen
    1.14.99.– Miscellaneous (requires further characterization)
1.15.–.– Acting on superoxide radicals as acceptor
1.16.–.– Oxidizing metal ions
    1.16. 1.– With NAD or NADP as acceptor
    1.16. 3.– With oxygen as acceptor
1.17.–.– Acting on $–CH_2–$ groups
    1.17. 1.– With NAD or NADP as acceptor
    1.17. 3.– With oxygen as acceptor
    1.17. 4.– With a disulfide as acceptor
    1.17.99.– With other acceptors
1.18.–.– Acting on reduced ferredoxin as donor
    1.18. 1.– With NAD or NADP as acceptor

continued

Table 1 – *continued*

	1.18. 6.– With dinitrogen as acceptor
	1.18.99.– With H^+ as acceptor
1.19.–.– Acting on reduced flavodoxin as donor	
	1.19. 6.– With dinitrogen as acceptor
1.97.–.– Other oxidoreductases	

Note: The Nomenclature Committee has accepted a recommendation that nicotinamide-adenine dinucleotide and nicotinamide-adenine dinucleotide phosphate should be abbreviated to NAD and NADP, respectively, rather than NAD^+ and $NADP^+$, as used previously. In addition to avoiding the erroneous implication that these two compounds will be positively charged at physiological pH values, the reasons for this are given in the Newsletter (1996). The reduced forms of these coenzymes are written as $NADH_2$ and $NADPH_2$, rather than NADH and NADPH.

The systematic name is in the form *donor:acceptor grouptransferase*. The recommended names are normally formed according to *acceptor grouptransferase* or *donor grouptransferase*.

Sometimes transferase reactions can be considered in different ways; for example, the general reaction shown above may be regarded as a transfer of the group Y from X to Z, and would therefore be termed a Y-transferase. However, it could also be considered as a breaking of the X–Y bond by the introduction of Z. For example, where Z represents phosphate, the process is often referred to as phosphorolysis and the enzyme catalysing the reaction as a *phosphorylase*. For systematic purposes, these enzymes are classified as *phosphotransferases*.

The aminotransferase (transaminase) reactions involve the transfer of a $-NH_2$ group and H to a compound containing a carbonyl group, in exchange for the $=O$ of that group (reaction [III]).

$$\mathrm{R1{-}CHNH_2{-}R2 + R3{-}CO{-}R4 = R1{-}CO{-}R2 + R3{-}CHNH_2{-}R4} \qquad \text{[III]}$$

Thus, the reaction could be regarded as being an oxidative deamination of the donor (e.g. an amino acid) linked to the reductive amination of the acceptor (e.g. oxo acid). Therefore these enzymes might be classified as oxidoreductases. However, since the unique distinctive feature of the reaction is the transfer of the amino group, these enzymes are classified as amino transferases (subclass 2.6.1.)

The second figure in the code number of the transferases denotes the general nature of the group transferred (a one-carbon group, 2.1; aldehydic or ketonic group, 2.2; acyl group, 2.3, etc.) and the third number further specifies that group (methyltransferase, 2.1.1; formyltransferase, 2.1.2, etc.). The exception is the case of the enzymes transferring phosphorus-containing groups (subclass 2.7), where the third number specifies the nature of the acceptor group.

Table 2 summarizes the structure of Class 2.

Example

EC 2.1.1.114

Recommended name: hexaprenyldihydroxybenzoate methyltransferase

Reaction: *S*-adenosyl-L-methionine + 3-hexaprenyl-4,5-dihydroxybenzoate = *S*-adenosyl-L-homocysteine + 3-hexaprenyl-4-hydroxy-5-methoxybenzoate

Other name(s): 3,4-dihydroxy-5-hexaprenylbenzoate methyltransferase; dihydroxyhexaprenylbenzoate methyltransferase

Systematic name: *S*-adenosyl-L-methionine:3-hexaprenyl-4,5-dihydroxylate *O*-methyltransferase

Comments: Involved in the pathway of ubiquinone synthesis. This enzyme has been listed as EC 2.1.1.64 in some sequence databases; but that enzyme catalyses a different reaction

Class 3. –. –. – Hydrolases

These enzymes catalyse the hydrolytic cleavage of bonds such as C–O, C–N, C–C and some other bonds, including phosphoric anhydride bonds. The overlapping specificities of many of these enzymes make it difficult to formulate general rules that are applicable to all members of this class. The systematic name usually takes the form *substrate X-hydrolase*, where X is the group removed by hydrolysis. The recommended name is, in many cases, formed by the name of the substrate with the suffix *-ase*. It is understood that the name of the substrate with this suffix means a hydrolytic enzyme.

Hydrolytic enzymes might be classified as transferases, since hydrolysis itself can be regarded as transfer of a specific group to water as the acceptor. Yet, in most cases, the reaction with water as the acceptor was discovered earlier and is considered as the main physiological function of the enzyme. This is why such enzymes are classified as hydrolases rather than as transferases.

The second number indicates the nature of the bond hydrolysed, and the third normally specifies the nature of the substrate, e.g. in the esterases the *carboxylic ester* *hydrolases* (3.1.1), *thiolester hydrolases* (3.1.2), *phosphoric monoester hydrolases* (3.1.3); in the glycosidases, the *O-glycosidases* (3.2.1), *N-glycosidases* (3.2.2), and so on.

The peptidases (formerly called peptide hydrolases; Class 3.4.–) cannot be accommodated within this general scheme. It is not even possible to give unambiguous systematic names because of variable specificities and great similarities between the actions of different peptidases. The enzymes are grouped into two sets of sub-subclasses, the endopeptidases (3.4.21–24 and 3.4.99) and exopeptidases (3.4.11–19), with the third figure also depending on the catalytic mechanism. A complete list of the peptidases is now available on-line (see *Peptidases*, 1998 in Further Reading section).

Subclasses 3.9 - 3.11, which each contain only one or two known enzymes, only contain one sub-subclass each, denoted by the figure 1 (e.g. 3.9.1.1, 3.10.1, 3.10.1.2 etc).

Table 3 summarizes the structure of Class 3.

Table 2 Class 2.–.–.– Transferases

2. 1. –.–	Transferring one-carbon groups
2. 1. 1. –	Methyltransferases
2. 1. 2. –	Hydroxymethyl-, formyl- and related transferases
2. 1. 3. –	Carboxyl- and carbamoyltransferases
2. 1. 4. –	Amidinotransferases
2. 2. –.–	Transferring aldehyde or ketone residues
2. 2. 1. –	a single subclass containing the transaldolases and transketolases
2. 3. –.–	Acyltransferases
2. 3. 1. –	Acyltransferases
2. 3. 2. –	Aminoacyltransferases
2. 4. –.–	Glycosyltransferases
2. 4. 1. –	Hexosyltransferases
2. 4. 2. –	Pentosyltransferases
2. 4.99. –	Transferring other glycosyl groups
2. 5. –.–	Transferring alkyl or aryl groups, other than methyl groups
2. 5. 1. –	A single subclass that includes a rather mixed group of such enzymes
2. 6. –.–	Transferring nitrogenous groups
2. 6. 1. –	Transaminases (aminotransferases)
2. 6. 3. –	Oximinotransferases
2. 6.99. –	Transferring other nitrogenous groups
2. 7. –.–	Transferring phosphorus-containing groups
2. 7. 1. –	Phosphotransferases with an alcohol group as acceptor
2. 7. 2. –	Phosphotransferases with a carboxyl group as acceptor
2. 7. 3. –	Phosphotransferases with a nitrogenous group as acceptor
2. 7. 4. –	Phosphotransferases with a phosphate group as acceptor
2. 7. 6. –	Diphosphotransferases
2. 7. 7. –	Nucleotidyltransferases
2. 7. 8. –	Transferases for other substituted phosphate groups
2. 7. 9. –	Phosphotransferases with paired acceptors
2. 8. –.–	Transferring sulfur-containing groups
2. 8. 1. –	Sulfurtransferases
2. 8. 2. –	Sulfotransferases
2. 8. 3. –	CoA-transferases
2. 9. –.–	Transferring selenium-containing groups

Examples

EC 3.1.2.23

Recommended name: 4-hydroxybenzoyl-CoA thioesterase

Reaction: 4-hydroxybenzoyl-CoA + H_2O = 4-hydroxybenzoate + CoA

Systematic name: 4-hydroxybenzoyl-CoA hydrolase

Comments: This enzyme is part of the bacterial 2,4-dichlorobenzoate degradation pathway

EC 3.4.22.38

Recommended name: cathepsin K

Reaction: Broad proteolytic activity. With small-molecule substrates and inhibitors, the major determinant of specificity is P2, which is preferably Leu, Met > Phe, and not Arg

Other names: cathepsin O and cathepsin X (both misleading, having been used for other enzymes); Cathepsin O_2

Comments: Prominently expressed in mammalian osteoclasts, and believed to play a role in bone resorption. In peptidase family C1

NOTE: The specificity of peptidases may be described in terms of a sequence of amino acid residues on either side of the peptide bond that is cleaved (scissile bond), which is indicated by

N-terminus

$- - - - P3 - P2 - P1P1' - P2' - P3' -$ C-terminus

Class 4. –. –. – Lyases

These enzymes cleave C–C, C–O, C–N and other bonds by means other than hydrolysis or oxidation. They differ from other enzymes in that two substrates are involved in one reaction direction but only one in the other. When acting on the single substrate, a molecule is eliminated leaving double bonds or rings. The systematic name is formed according to the pattern *substrate group-lyase*. The hyphen is an important part of the name and, to avoid confusion, should not be omitted, e.g. *hydro-lyase* not 'hydrolyase'. In the recommended names, expressions like *decarboxylase* or *aldolase* (in case of elimination of CO_2 or aldehyde, respectively) are used. *Dehydratase* is used for those enzymes catalysing the elimination of water. In cases where the reverse reaction is much more important, or the only one demonstrated, *synthase* (not synthetase) may be used in the name. Although the term SYNTHETASE has

Table 3 Class 3.–.–.–. Hydrolases

3. 1. –.– Acting on ester bonds	
	3.1. 1. – Carboxylic ester hydrolases
	3.1. 2. – Thiolester hydrolases
	3.1. 3. – Phosphoric monoester hydrolases
	3.1. 4. – Phosphoric diester hydrolases
	3.1. 5. – Triphosphoric monoester hydrolases
	3.1. 6. – Sulfuric ester hydrolases
	3.1. 7. – Diphosphoric monoester hydrolases
	3.1. 8. – Phosphoric triester hydrolases
	3.1.11. – Exodeoxyribonucleases producing 5′-phosphomonoesters
	3.1.13. – Exoribonucleases producing 5′-phosphomonoesters
	3.1.14. – Exoribonucleases producing other than 5′-phosphomonoesters
	3.1.15. – Exonucleases active with either ribo- or deoxyribonucleic acids and producing 5′-phosphomonoesters
	3.1.16. – Exonucleases active with either ribo- or deoxyribonucleic acids and producing other than 5′-phosphomonoesters
	3.1.21. – Endodeoxyribonucleases producing 5′-phosphomonoesters
	3.1.22. – Endodeoxyribonucleases producing other than 5′-phosphomonoesters
	3.1.25. – Site-specific endodeoxyribonucleases specific for altered bases
	3.1.26. – Endoribonucleases producing 5′-phosphomonoesters
	3.1.27. – Endoribonucleases producing other than 5′-phosphomonoesters
	3.1.30. – Endonucleases active with either ribo- or deoxyribonucleic acid and producing 5′-phosphomonoesters
	3.1.31. – Endonucleases active with either ribo- or deoxyribonucleic acid and producing other than 5′-phosphomonoesters
3. 2. –.– Glycosidases	
	3.2. 1. – Hydrolysing *O*-glycosyl compounds
	3.2. 2. – Hydrolysing *N*-glycosyl compounds
	3.2. 3. – Hydrolysing *S*-glycosyl compounds
3. 3. –.– Acting on ether bonds	
	3.3. 1. – Thioether hydrolases
	3.3. 2. – Ether hydrolases
3. 4. –.– Acting on peptide bonds (peptidase)	
	3.4.11. – Aminopeptidases
	3.4.13. – Dipeptidases
	3.4.14. – Dipeptidyl-peptidases and tripeptidyl-peptidases
	3.4.15. – Peptidyl-dipeptidases
	3.4.16. – Serine-type carboxypeptidases
	3.4.17. – Metallocarboxypeptidases
	3.4.18. – Cysteine-type carboxypeptidases
	3.4.19. – Omega peptidases
	3.4.21. – Serine endopeptidases
	3.4.22. – Cysteine endopeptidases
	3.4.23. – Aspartic endopeptidases
	3.4.24. – Metalloendopeptidases
	3.4.99. – Endopeptidases of unknown catalytic mechanism
3. 5. –.– Acting on carbon–nitrogen bonds, other than peptide bonds	
	3.5. 1. – In linear amides
	3.5. 2. – In cyclic amides
	3.5. 3. – In linear amidines
	3.5. 4. – In cyclic amidines
	3.5. 5. – In nitriles
	3.5.99. – In other compounds

continued

Table 3 – *continued*

3. 6. –.–	Acting on acid anhydrides
3.6. 1. –	In phosphorus-containing anhydrides
3.6. 2. –	In sulfonyl-containing anhydrides
3. 7. –.–	Acting on carbon–carbon bonds
3.7. 1. –	In ketonic substances
3. 8. –.–	Acting on halide bonds
3.8. 1. –	In C-halide compounds
3. 9. –.–	Acting on phosphorus–nitrogen bonds
3.10. –.–	Acting on sulfur–nitrogen bonds
3.11. –.–	Acting on carbon–phosphorus bonds
3.12. –.–	Acting on sulfur–sulfur bonds

sometimes been used in the names of enzymes from this class, that use is discouraged in order to prevent confusion with enzymes from Class 6 (see below).

Various subclasses of the lyases include pyridoxal-phosphate enzymes that catalyse the elimination of a β- or γ-substituent from an α-amino acid, followed by a replacement of this substituent by some other group. In the overall replacement reaction, no unsaturated end product is formed; therefore, these enzymes might formally be classified as *alkyltransferases* (EC 2.5.1.-). However, there is ample evidence that the replacement is a two-step reaction involving the transient formation of enzyme-bound α,β- (or β,γ-)unsaturated amino acids. According to the rule that the first reaction is indicative for classification, these enzymes are correctly classified as *lyases*. Examples are *tryptophan synthase* (EC 4.2.1.20) and *cystathionine β-lyase* (EC 4.2.1.22). The second figure in the code number indicates the bond broken: 4.1 are carbon–carbon lyases, 4.2 are carbon–oxygen lyases, and so on. The third figure gives further information on the group eliminated (e.g. CO_2 in 4.1.1 and H_2O in 4.2.1).

Table 4 Class 4. –.–.– Lyases

4. 1. –.–	Carbon–carbon lyases
4. 1. 1 –	Carboxy-lyases
4. 1. 2 –	Aldehyde-lyases
4. 1. 3 –	Oxo-acid-lyases
4. 1.99 –	Other carbon–carbon lyases
4. 2. –.–	Carbon–oxygen lyases
4. 2. 1 –	Hydro-lyases
4. 2. 2 –	Acting on polysaccharides
4. 2.99 –	Other carbon–oxygen lyases
4. 3. –.–	Carbon–nitrogen lyases
4. 3. 1 –	Ammonia-lyases
4. 3. 2 –	Amidine-lyases
4. 3. 3 –	Amine-lyases
4. 3.99 –	Other carbon–nitrogen-lyases
4. 4. –.–	Carbon–sulfur lyases
4. 5. –.–	Carbon–halide lyases
4. 6. –.–	Phosphorus-oxygen lyases
4.99. –.–	Other lyases

Table 4 summarizes the structure of Class 4.

Example

EC 4.1.2.39

Recommended name: hydroxynitrilase

Reaction: 2-hydroxyisobutyronitrile = cyanide + acetone

Other name(s): hydroxynitrile lyase; oxynitrilase

Systematic name: 2-hydroxyisobutyronitrile acetone-lyase

Comments: The enzyme from *Hevea* (rubber tree) and *Manihot* spp. (cassava) accepts aliphatic and aromatic hydroxynitriles, unlike EC 4.1.2.11, which does not act on aliphatic hydroxynitriles. 2-Hydroxyisobutyronitrile (acetone cyanohydrin) is liberated by glycosidase action on linamarin

Class 5. –. –.– Isomerases

These enzymes catalyse geometric or structural changes within one molecule. According to the type of isomerism involved they may be called *racemases*, *epimerases*, *cis–trans-isomerases*, *isomerases*, *tautomerases*, *mutases* or *cycloisomerases*. The second number denotes the type of isomerism involved, and the third number the type of substrate. In some cases, the reaction involves an intermolecular oxidoreduction, but since the donor and acceptor groups are in the same molecule they are classified as isomerases rather than oxidoreductases, even though they may contain firmly bound NAD or NADP.

Table 5 summarizes the structure of Class 5.

Example:

EC 5.1.99.4

Recommended name: α-methylacyl-CoA racemase

Table 5 Class 5. –.–.– Isomerases

5. 1. –.– Racemases and epimerases	
	5. 1. 1. – Acting on amino acids and derivatives
	5. 1. 2. – Acting on hydroxy acids and derivatives
	5. 1. 3. – Acting on carbohydrates and derivatives
	5. 1.99. – Acting on other compounds
5. 2. –.– *cis–trans*-Isomerases	
5. 3. –.– Intramolecular oxidoreductases	
	5. 3. 1. – Interconverting aldoses and ketoses
	5. 3. 2. – Interconverting keto- and enol-groups
	5. 3. 3. – Transposing C=C bonds
	5. 3. 4. – Transposing S–S bonds
	5. 3.99. – Other intramolecular oxidoreductases
5. 4. –.– Intramolecular transferases (mutases)	
	5. 4. 1. – Transferring acyl groups
	5. 4. 2. – Phosphotransferases (phosphomutases)
	5. 4. 3. – Transferring amino groups
	5. 4.99. – Transferring other groups
5. 5. –.– Intramolecular lyases	
5.99. –.– Other isomerases	

Reaction: (2*S*)-2-methylacyl-CoA = (2*R*)-2-methylacyl-CoA

Systematic name: 2-methylacyl-CoA 2-epimerase

Comments: α-methyl-branched acyl-CoA derivatives with chain lengths of more than C_{10} are substrates. Also active towards some aromatic compounds (e.g. ibuprofen) and bile acid intermediates, such as trihydroxycoprostanoyl-CoA. Not active towards free acids

Class 6. –. –.– Ligases

These enzymes catalyse the joining together (ligating) of two molecules with the concomitant hydrolysis of a diphosphate bond in ATP or a similar triphosphate. The systematic enzyme name takes the form *A:B ligase* (*XDP- or XMP-forming*). The recommended name often takes the form A–B ligase. Sometimes the name *synthase* is used for the recommended name to emphasize the synthetic nature of the reaction, which can also be helpful if the reaction is complex. The name synthetase is also sometimes used instead of synthase in the names of enzymes in this class.

The second figure in the code number indicates the bond formed: 6.1 for C–O bonds (enzymes acylating tRNA), 6.2 for C–S bonds (acyl-CoA derivatives), etc. Sub-subclasses are only in use in the C–N ligases (6.3), which include the amide synthases (6.3.1), the peptide synthases (6.3.2), enzymes forming heterocyclic rings (6.3.3), etc.

Table 6 summarizes the structure of Class 6.

Example

EC 6.2.1.33

Recommended name: 4-chlorobenzoate-CoA ligase

Reaction: 4-chlorobenzoate + CoA + ATP = 4-chlorobenzoyl-CoA + AMP + diphosphate

Systematic name: 4-chlorobenzoate : CoA ligase

Comments: Requires Mg^{2+}. This enzyme is part of the bacterial 2,4-dichlorobenzoate degradation pathway.

Limitations and Problems

Isoenzymes may not be easily accommodated in any system of classification simply in terms of reaction catalysed. For example, there are about 20 different isoenzymes of alcohol dehydrogenase in human liver. These have been organized into broad classes in terms of their electrophoretic mobilities and, more precisely, in terms of their sequences and genetic origin. These classes show very different chain-length specificities for primary aliphatic alcohols and also different inhibitor specificities. However, since they all oxidize primary alcohols and have a strong preference towards NAD as the coenzyme, they are all grouped together under the general heading of EC 1.1.1.1. Furthermore, problems also arise from species differences; for example, class EC 1.1.1.1 includes NAD-dependent alcohol dehydrogenases from all species, although the mammalian liver and yeast enzymes, for example, are profoundly different in structure and behaviour. Only when isoenzymes have very different substrate specificities might classification by function provide the whole solution. For example, liver glucokinase is now recognized to be a member of the hexokinase family of isoenzymes (hexokinase type IV) and is classified as a hexokinase (EC 2.7.1.1), whereas the name glucokinase

Table 6 Class 6. –.–.– Ligases

6. 1. –.– Forming carbon–oxygen bonds	
	6. 1. 1. – Ligases forming aminoacyl-tRNA and related compounds
6. 2. –.– Forming carbon–sulfur bonds	
	6. 2. 1. – Acid–thiol ligases
6. 3. –.– Forming carbon–nitrogen bonds	
	6. 3. 1. – Acid–ammonia (or amine) ligases (amide synthases)
	6. 3. 2. – Acid–amino-acid ligases (peptide synthases)
	6. 3. 3. – Cyclo-ligases
	6. 3. 4. – Other carbon–nitrogen ligases
	6. 3. 5. – Carbon–nitrogen ligases with glutamine as amido-*N*-donor
6. 4. –.– Forming carbon–carbon bonds	
6. 5. –.– Forming phosphoric ester bonds	

(EC 2.7.1.2) is specifically recommended for the enzyme from invertebrates and microorganisms that has a high specificity for glucose. In other cases, this problem is being addressed by linking the electronic form of the enzyme list to other appropriate databases, based on structural considerations.

Information and Updates

New enzymes and new functions of existing enzymes are being discovered at a rapid pace and work on revising and expanding the list of enzymes is a continuing operation. Suggestions for enzymes that should be included, or for revisions and corrections to existing entries can be submitted electronically using forms available through the IUBMB Nomenclature Committee Enzyme Nomenclature or SWISSPROT ENZYME home pages. Alternatively, material for all enzyme classes except the peptidases (Class 3.4. –.–) can be sent by e-mail or regular mail to Dr Sinead Boyce (Department of Biochemistry, Trinity College, Dublin 2, Ireland; E-mail: sboyce@tcd.ie). Material relating to the peptidases should be sent to Dr Alan J. Barrett (Peptidase Laboratory, Department of Immunulogy, Babraham Institute, Babraham CB2 4AT, UK; E-mail; alan.barrett@bbsrc.ac.uk). After these have been checked and considered by the Nomenclature Committee as a whole, they are incorporated into the *Enzyme Nomenclature* database, which is being prepared for on-line circulation.

Further Reading

Alberty RA, Cornish-Bowden A, Gibson QH *et al.* (1996) Recommendations for nomenclature and tables in biochemical thermodynamics. *European Journal of Biochemistry* **240**: 1–14.

Dixon M and Webb EC (1958) *Enzymes*, pp. 183–227. London: Longmans Green.

Enzyme Nomenclature (1992) Recommendations of the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology on the Nomenclature and Classification of Enzymes. New York: Academic Press.

Enzyme Supplements (1999) Prepared for the NC-IUBMB by Tipton KF and Boyce S. [http://www.chem.qmw.ac.uk/iubmb/enzyme/]

IUBMB Nomenclature Committee Enzyme Nomenclature. Recommendations of the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology. [http://www.chem.qmw.ac.uk/iubmb/enzyme/]

Newsletter (1996) of the IUPAC-IUBMB Joint Commission on Biochemical Nomenclature (JCBN) and Nomenclature Committee of IUBMB (NC-IUBMB). *Archives of Biochemistry and Biophysics* (1997) **344**: 242–252; *Biochemical Journal* (1997) **327**: 311–319; *Chemistry International* (1997) **19**: 116–119; *European Journal of Biochemistry* (1997) **247**: 733–739; *Glycoconjugate Journal* (1998) **15**: 637–647. Also available on-line at [http://www.chem.qmw.ac.uk/iubmb/newsletter/]

Panico R, Richer J-C Powell WH (1994) *A Guide to IUPAC Nomenclature of Organic Compounds*. Oxford: Blackwell Science.

Peptidases (1998) Prepared for the NC-IUBMB by Barrett AJ, Bond JS, Fiedler F *et al.* [http://www.chem.qmw.ac.uk/iubmb/enzyme/EC34/]

SWISSPROT ENZYME. Swiss Institute of Bioinformatics (SIB) Enzyme nomenclature database primarily based on the recommendations of the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology (IUBMB) [http://www.expasy.ch/enzyme]